RULES AND REGULATIONS FOR THE WRITTEN EXAMS

AT THE FACULTY OF ENGINEERING AND NATURAL SCIENCES

Please read this document carefully, sign the bottom part, and include this form with your solution sheet.

- The exam duration (start-end times) will be written on your exam sheet or will be announced, you should adhere to these times.
- You cannot leave the examination room in the **first 30 minutes** of the exam.
- You are **NOT allowed** to take the exam once a student has left one of the exam rooms.
- Talking with another student during the exam for any reason (asking for an explanation, for a calculator or eraser or any other object) is **NOT allowed**.
- If you **GIVE** to or **RECEIVE** from a student any item (eraser, calculator, pen, etc.) during the exam, this will be taken as a sign of cheating.
- **TURN OFF** your **CELLPHONES AND SMART DEVICES**; and PLACE THEM **away from you** such that those **CANNOT BE SEEN** from outside.
- **HAVING a CELLPHONE or a SMART DEVICE** that is VISIBLE FROM OUTSIDE during the exam is **NOT ALLOWED**. Interacting with a cellphone in any way (EVEN IF IT IS TURNED OFF) will be taken as a sign of cheating.

In case of a sign of cheating, or if you do not obey the above rules, there will be a disciplinary action taken towards you based on the STUDENT DISCIPLINARY REGULATIONS of YÖK Items 2.5, 2.7, and 2.8.

I understand the above-mentioned rules and regulations, and I hereby accept them, and any consequences following my actions during the exam.

Student Full Name:

Student Number:

Course Code and Title:

AIN-2002 INTRODUCTION TO DATA SCIENCE

Signature:

Date:

INTRODUCTION TO DATA SCIENCE

MIDTERM EXAM

(This is a 100-minute exam)

Q.1:	Q.2:	Q.3:	Q.4:	TOTAL:

Q.1) Please conduct necessary R code for the following expressions:

a. Create a function in R that returns the square of a number (4pts)

b. Plot the function (x) = $x^2 + 3x + 2$ for x in [-10, 10] (4pts)

c. Simulate tossing a coin 10 times (4pts)

d. Find the mean and median of 2, 4, 6 and 8 (4pts)

e. Create a boxplot of random normal data (4pts)

Q.2) (20) You are a data analyst for a retail company and are tasked with analysing whether a new advertising campaign has significantly impacted sales. The company claims that the average sales per day before the campaign was $500. After running the campaign for 30 days, you collect the following data:

- Sample mean of daily sales during the campaign: $550
- Sample standard deviation: $100
- Sample size: 30 days

According to the information given, please proceed with the following:

a. Define Null and Alternative Hypothesis (5pts)
b. Perform t-test thru 0.05 significance level (5pts)
c. Define degrees-of-freedom (**df**) (5pts)
d. Interpret the results. (If the campaign had significant effect on sales or not) (5pts)

Q.3) (20pts) The dataset presented in the exhibit displays stress levels in relation to daily social media usage among young adults aged **18 to 25 years**. The **second column** contains the observed (measured) stress levels collected from participants. And the **third column** presents the predicted stress levels generated by a regression model.

social media usage (hrs/day)	measured stress level (likert scale)	predicted stress level (likert scale)
0.5	3.0	2.9
1.0	3.5	3.4
1.5	4.0	4.1
2.0	4.5	4.7
2.5	5.0	4.9
3.0	5.5	5.3
4.0	6.5	8.0
5.0	6.5	10.2
6.0	7.0	13.0
7.0	7.5	21.0
8.0	8.0	24.0

Using the data provided, answer the following questions:

a. Calculate the Root Mean Squared Error (RMSE) between the measured and predicted stress levels. (2pts)

b. Compute the coefficient of determination (R^2) to assess the model's goodness of fit. (3pts)

c. What does the RMSE value reveal about the accuracy of the model's predictions? What would be the expected range for RMSE? (5pts)

d. How does the R^2 value reflect the model's performance? What would be an expected range for RMSE? (5pts)

e. Observing the predicted values, does the model tend to overestimate or underestimate stress levels at higher social media usage? Justify your answer. (5pts)

Q.4) (20 pts) Given the information above, provides the gathered data on the use of Helium-3 (^3He) – a rare earth element as a fuel source for a spaceship, including the distance it has enabled the spacecraft to travel.

Helium-3 (^3He) (kg)	distance travelled (km)
1	1.5
2	2.5
3	5
4	9
5	16
6	29
7	52
8	95
9	170
10	310

Please answer the questions below according to the data collected:

a. Please determine the dependent and independent variables (3 pts)

b. Is the relationship between dependent and independent variables linear or non-linear? (3 pts)
(Please elaborate based on what information you are deciding)

c. Determine the type of regression between dependent and independent variables (5 pts)
(TIP: plot the data, observe the trend)

d. It is comforting to transform non-linear trends into linear trends. Does this apply on the example dataset or not? If yes, how? (5 pts)

e. Determine and express the regression equation. (5 pts)

f. Find the Intercept (β_0) and Slope (β_1) (5 pts)

g. Express the model (9 pts)

h. Based on the model calculate how many kilometres the spaceship can travel given 29 kilograms of Helium-3 (^3He)? (5 pts)

Formulas

Sum of squared residuals	$\mathrm{SS}_{\mathrm{res}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$
Total sum of squares	$\mathrm{SS}_{\mathrm{tot}} = \sum_{i=1}^{n} (y_i - \bar{y})^2$
Coefficient of determination (R^2)	$R^2 = 1 - \frac{\mathrm{SS}_{\mathrm{res}}}{\mathrm{SS}_{\mathrm{tot}}}$
Simple linear regression	$y = \beta_0 + \beta_1 x$
Multiple linear regression	$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$
Exponential regression	$y = ae^{bx}$
2nd degree Polynomial reg.	$y = \beta_0 + \beta_1 x + \beta_2 x^2$
Intercept	$\beta_0 = \bar{y} - \beta_1 \bar{x}$
Slope	$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$
Logarithmic equation	$\ln(ae^{bx}) = \ln a + bx$ $\ln(x) = y$ if and only if $e^y = x$
Root Mean Square Error (RMSE)	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$
t-statistic	$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$
Degrees of freedom	$\mathrm{df} = n - 1$

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										